Anirudh Tiwary

201-744-5517 | anitiwary87@gmail.com | github.com/ani-tiwary | linkedin.com/in/ani-tiwary

Education

Rutgers University — New Brunswick, NJ

B.S. Mathematics, Computer Science, and Philosophy | GPA: 3.85 — *2024 — 2027*

- Coursework: Computer Architecture, Algorithms, Systems Programming, Linear Optimization, Calculus IV
- Involvement: Rutgers Agentic Intelligence Lab Lead, Rutgers Economics Labs Analyst Team Lead, Midway Chapter Technical Lead, USACS Technical Blog Writer, Rutgers Chess Team Player

Academy for Information Technology — Scotch Plains, NJ

High School Diploma – Computer Science Track | SAT: 1570 — *2020 — 2024*

- Coursework: Mathematical Statistics & Data Science, Advanced Software Development

Experience

Goldman Sachs — Software Engineer Intern — Jun. 2026 — Aug. 2026

New York, NY

- Built internal MCP server for aggregation and analysis of **over 1M** liquidity risk data points across **Snowflake**, **AWS S3**, and proprietary data lakes
- Engineered LLM context & retrieval layer to automate **∼75%** of data pipeline for **hundreds** of risk managers worldwide

Scarlet Sync — Software Engineer — Apr. 2025 — Present

Rutgers University — *New Brunswick, NJ*

- Co-founded and scaled (**1000+ users**) Rutgers' first course scheduling platform featuring conflict detection, automated registration, and collaborative tools.
- Architected production-ready **Next.js + Supabase** system generating **5K+ schedules in under 2s** with secure **OAuth**-based authentication.
- Implemented **Redis-based caching and Pub/Sub** for real-time seat updates and query acceleration, enabling low-latency scheduling performance.

Lykke (Startup) — Software Engineer — Jan. 2026 — Feb. 2026

Remote — *San Francisco, CA*

- Built a **multi-modal RAG pipeline** to ingest and structure **10GB+** of high-velocity, multi-modal data (text, documents, voice) for a conversational AI application.
- Architected data integration pipelines to unify disparate sources into a **domain-specific knowledge graph**, enabling structured querying and intelligent retrieval.

Aresty Research Program — ML Research Asst. — May 2025 — Present

Rutgers University — *New Brunswick, NJ*

- Built a **machine learning pipeline** using **Python, Pandas, Scikit-learn** to predict railway track degradation, achieving **73% accuracy** with Ridge Regression.
- Engineered **19 spatial-temporal features** and applied **chronological validation** to prevent data leakage.
- Enabled **predictive maintenance insights** with estimated **25–30% cost reduction potential** through data-driven modeling.

Rutgers Economics Labs (REL) — Research Analyst & AI Lead — Sep. 2024 — Present

Rutgers University — *New Brunswick, NJ*

- Performed **data science analysis for NJ Department of Labor (NJDOL) and Federal HUD**, building **cohort-based population and labor supply/demand models** to generate projections (2022–2050) and identify **regional labor gaps** for policy and infrastructure planning.
- Designed and contributed to **RAIL (Rutgers Agentic Intelligence Lab)**, a modular multi-model system for ingesting, cleaning, and structuring large economic data into reusable, queryable pipelines.

Technical Skills

Languages: Java, Python, C, JavaScript, HTML/CSS, TypeScript, SQL, GDScript
ML & Data Tools: Language Models, Pandas, Scikit-learn, TensorFlow, NumPy, Seaborn
Frameworks & Platforms: Supabase, Next.js, React, Tailwind CSS, Redis, Flask, Node.js
Other Tools: Linux, Git, OAuth, Android Studio

Accolades

- Placed **3/500+** in ASA Data Challenge using **Python (Pandas, Scikit-learn)** for regression and inference.
- **Track winner** at Rutgers Health Hacks for building an ML-based yoga pose estimation system using **computer vision**.
- Won **3rd** in Rutgers Shark Tank pitching **Scarlet Sync**, demonstrating technical market knowledge and communication.